



HEANG Denis

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EDUCATION

Institute of Technology of Cambodia

Expected Graduation: 2027

Udemy Full Stack Development Bootcamp

Relevant Coursework: Data Structures, Algorithms, Database Systems, Machine Learning, Linear Algebra, Probability & Statistics.

TECHNICAL SKILLS

Programming: Python, R, PostgreSQL, SQL, JavaScript

Machine Learning/Data Science: Scikit-learn, LLM, Pandas, NumPy, Matplotlib, Seaborn, Tensorflow, Scipy, Langchain.

Frameworks: FastAPI, Django, React, Node.js

Tools: Git, Hugging Face, VScode, Power BI, Excel, Jupyter Notebook.

LANGUAGES

Khmer (Native), **English** (B2, Upper-Intermediate), **French** (A2, Elementary)

WORK EXPERIENCES

Scientific Day

Phnom Penh

Project Showcase **Rain prediction** | **Python**(Scikit-learn, Matplotlib, Pandas, Django)

July 2025

- Developed a **rain prediction model using Scikit-learn** based on historical weather data in Cambodia.
- Performed **data cleaning and preprocessing using Pandas** to prepare datasets for machine learning.
- Visualized weather patterns and prediction results using **Matplotlib** to improve model interpretation.
- Developed a simple **Django web interface** allowing users to input weather conditions and receive rainfall predictions.

SCHOOL / PERSONAL PROJECTS

Credit Risk Analysis, Prediction & RAG Assistant | **Power BI, Python** (FastAPI, scikit-learn, Pandas, NumPy), **Llama 3.2** (1B & 3B), LangChain, ChromaDB, Whisper, React.

- Processed **30,000+** loan records for credit risk model (EDA, cleaning, feature engineering) using **Pandas NumPy, Power Bi**.
- Trained **Logistic Regression, Neural Network, CatBoost** models with **Scikit-learn** to predict probability of default **Basel-aligned risk grades (A–F)** with **Waterfall Chart**.
- Built Retrieval-Augment Generation (RAG) chatbot using **Llama 3.2 (1B, 3B)** powered by **Meta, LangChain, and ChromaDB** vector store to answer complex questions grounded in loan documents and risk policies.
- Integrated a **Whisper-medium** model for speak-to-text in English.
- Implemented a **FastAPI backend** and connected it to a **React frontend** for real-time voice input, predictions and chat interface.

House Price Prediction Web App | **Python**(Beautiful Soup, PyTorch, Streamlit, Pandas, NumPy, Scikit-learn, Matplotlib).

- Collected **8,100+ Florida property listings** from Redfin using **BeautifulSoup**, building a dataset for housing price analysis.
- Built and deployed a **Streamlit web application** to predict whether a property.
- Developed a **PyTorch neural network model** with a full ML pipeline including data cleaning, IQR-based outlier removal.
- Engineering an **interactive data visualizations using Matplotlib** to analyze price ranges.

AI Educational Chatbot | **Python, Node.js, Express.js, Flask, PostgreSQL, LangChain, ChromaDB, Google Cloud**

- Built a **AI chatbot platform** using **Node.js, Express.js, Flask, and PostgreSQL**, enabling students to query curriculum content through conversational AI. Implemented a **Retrieval-Augmented Generation (RAG) pipeline** with **LangChain and ChromaDB** to retrieve relevant **Khmer literature documents** before LLM response generation.
- Designed and optimized a **PostgreSQL relational database** with schemas, triggers, and functions to manage **chat history, user sessions, and usage tracking**.

LEADERSHIP

AI Educational Chatbot Project, Team Lead (4 members)

- Led a team of 4, assigning roles across backend, frontend, and database development and coordinating project workflow.
- Collected, cleaned, and prepared Khmer text data for training and retrieval.
- Designed and implemented the model architecture for the chatbot system.
- Coordinated closely with team members to ensure on-time delivery and successful integration of all components.